Task-1: Research Paper Explanations

AI/ML Intern Assessment

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Paper Title: *Attention is All You Need*

Attention Is All You Need (Vaswani et al., 2017) is a groundbreaking paper by researchers at **Google Brain** and the **University of Toronto**, published on **arXiv** which introduced the Transformer architecture, a model that revolutionized natural language processing by proposing **attention mechanism** instead of recurrent (RNN) or convolutional (CNN) networks. **Sequential processing (RNN, CNN, LSTM, GRU etc.)** limited parallelization and made it difficult to capture long-range dependencies in text. The authors proposed that recurrence is not necessary. A model can learn relationships between words using self-attention which allows every word in a sentence to attend to every other word simultaneously. This design enables parallel processing of entire sequences, drastically reducing training time from weeks on RNNs to days while achieving state-of-the-art **BLEU** scores of **28.4** on **English-to-German WMT 2014** (outperforming previous bests by 2+ points) and **41.8** on **English-to-French**.

The Transformer is built from stacked encoder and decoder layers that use multi-head self-attention and feedforward networks. Since it does not process text sequentially, it adds positional encoding to maintain word order information. Self-attention assigns weights to show how strongly each word relates to others in the sentence. This helps the model understand context more effectively while enabling faster parallel training.

For example, in the sentence *"Shaikat thanked Shibly because he helped him,"* the model must determine who "he" refers to. Self-attention allows the model to compare all words in the sentence and assign stronger importance to the correct relationship. This ability to capture long-distance word dependencies helped the Transformer achieve state-of-the-art translation results and become the foundation of models like **BERT** and **GPT**.

<u>Paper Title: *Algorithm of Thoughts*</u>

"***Algorithm of Thoughts**: Enhancing Exploration of Ideas in Large Language Models*" is a 2023 paper by Bilgehan Sel et al., from **UC Berkeley** and **Salesforce AI Research**, published on **arXiv (2308.10379)** and accepted at **ICML 2024**. It is a reasoning framework designed to improve the problem-solving ability of large language models (**LLMs**).

It extends earlier methods like **Chain-of-Thought (CoT)** prompting and **Tree-of-Thought (ToT)** reasoning, where models are asked to think through a problem step by step. While Chain-of-Thought usually moves through reasoning in a straightforward and step-by-step manner, Algorithm of Thought improves on this by arranging the thinking process in a more organized and methodical way similar to carefully following a clear set of step-by-step instructions. Instead of generating loosely connected thoughts, the model is guided to move through carefully arranged steps that can be checked and verified along the way. This makes the reasoning process more structured, transparent, and reliable.

For example,

Problem:

If a car travels 60 km/h for 2 hours, how far does it travel?

Normal response:

- 60 × 2 = 120 km

Algorithm of Thought style:

1. Identify speed = 60 km/h
2. Identify time = 2 hours
3. Use formula: distance = speed × time
4. Compute: 60 × 2 = 120
5. Final answer: 120 km

It explicitly follows structured reasoning.

Thank You

For reviewing this document.

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